Numerical simulations for SDE's (Stochastic differential equations)

↳ random process that evolves in time

Simulation of SDE's and Langevin-type stochastic models:
Langevin-type models

diffusion

$$dx = f(x)dt + qdw$$

$f(x)$ is known as the drift
q is the diffusion coefficient

w = Brownian motion or diffusion

We write this in differential form, rather than

$$\frac{dx}{dt} = f(x) + q\frac{dw}{dt}$$

since w is continuous but not differentiable.

General Stochastic Differential Equations (SDE):

$$dx = A(x,t)dt + B(x,t)dw$$

The behavior of the increment dw suggests a simple approach for

Simulate at discrete time steps: $t_j = j\Delta t$

Note:
$\frac{dx}{dt} = f(x)$ is a deterministic dynamical system

Discrete approximation for the derivative:

$$\frac{dx}{dt} \approx \frac{x(t+\Delta t) - x(t)}{\Delta t}$$

Behavior of dw:

Then, $\mathrm{Var}[\Delta w] = \Delta t$ $\quad dw \approx \Delta w \sim N(0, \sqrt{\Delta t})$] this is derived below

$\sqrt{\Delta t}$: standard deviation

$$x(t+\Delta t) = x(t) + A(x(t),t)\,\Delta t + B(x(t),t)\,\sqrt{\Delta t} Z$$

where $Z \sim N(0,1)$. Recall: $\mathrm{Var}(cZ) = E[c^2(Z - E[Z])^2] = c^2 E[(Z - E[Z])^2] = c^2\,\mathrm{Var}[Z]$

In general, writing $x_j = x(j\Delta t)$ we have an iterative procedure to get x_j for all j, starting with an initial condition $x = x_0$ and computing to a time $T = K\Delta t$.

$x(1) = x_0$;
for $j = 1 : K-1$
$x_{j+1} = x_j + A(x_j, t_j)\,\Delta t + B(x_j, t_j)\sqrt{\Delta t} Z_j$
end

approximation for $dw \approx \Delta w$, the diffusion step

usual Euler method

Z_j are $K-1$ independent $N(0,1)$ random variables
This is the Euler-Maruyama method. It is generally associated with the Ito interpretation of SDE's (later). This has implications for the dynamics via the interpretation of the noise increments.

Higher order methods: Reference: Kloeden and Platen, Numerical Solution of SDE's, 1992. Both Weak (in distribution) and Strong (pathwise) methods.

Error for Euler-Maruyama method.

For higher dimensional SDE:

$$d\mathbf{x} = \mathbf{A}(\mathbf{x},t)dt + \mathbf{B}(\mathbf{x},t)d\mathbf{w} \qquad (16)$$

$\mathbf{x}, \mathbf{A}$ are d-dimensional, $\mathbf{w}$ is n-dimensional, B is $d \times n$ dimensional.

Example:

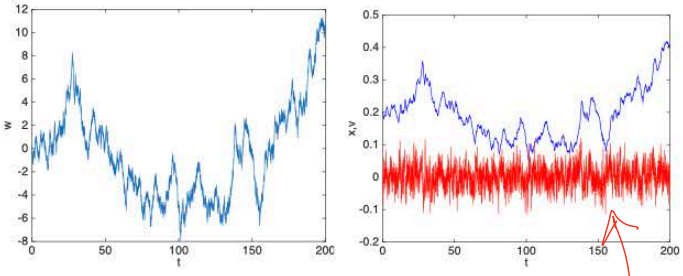
$$mx_{tt} = -\eta x_t + qX$$

η = damping constant

For X as above, we write (approximate w/ diffusion)

$$\begin{aligned} dx &= vdt \\ dv &= -\frac{\eta}{m}vdt + \frac{q}{m}dw \end{aligned}$$

Note: the drift is linear (in v), diffusion coefficient is constant
v is an Ornstein-Uhlenbeck (O-U) process - linear drift and constant diffusion coefficient



⇒ w behaves as diffusion w ~ N(0, t)

Figure 2: Realization for noise process, particle position, and velocity

the long time behavior of v is a Normal r.v.

$$\begin{aligned} &x(1) = x_0; v(1) = v_0 \\ &\text{for } j = 1 : K-1 \\ &x_{j+1} = v_j \Delta t \\ &v_{j+1} = v_j - \frac{\eta}{m} v_j \Delta t + \frac{q}{m}\sqrt{\Delta t} Z_j \\ &\text{end} \end{aligned}$$

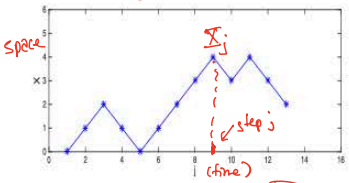
Note: v is an Ornstein-Uhlenbeck process (OU process). It has a linear drift term and a constant coefficient for the diffusion term. OU processes have a Gaussian stationary density with a constant mean and variance, i.e. normally distributed. That is, as $t \to \infty$, $v \sim N(0, q/\sqrt{2\eta})$.

3 Master and Forward equations

Random walk as a Markov process

How do we know that $w \sim N(0, \sqrt{t})$? w is not differentiable?

(17) We define X_n as a random walk, where X_n =position after n steps, with probabilities of taking a step up or down:



(at each time step)

P(step up: $X_{n+1} = X_n + 1) = p$

P(step down: $X_{n+1} = X_n - 1) = 1 - p$

X_{n+1} depends only on X_n

Each step is independent of previous steps (Markovian) (depends only on present location), so we can write $P(X_n = k)$ in terms of the conditional probability, conditioning on the location at the previous step:

(see Fact from conditional probability below)

$$\begin{aligned} P(X_n = k) &= P(X_n = k | X_{n-1} = k-1) P(X_{n-1} = k-1) \\ &+ P(X_n = k | X_{n-1} = k+1) P(X_{n-1} = k+1) \\ &= pP(X_{n-1} = k-1) + (1-p) P(X_{n-1} = k+1) \end{aligned}$$

only 2 options – up or down

In general,

$$P(X_n = k) = P(\text{ location after n steps, } m \text{ + steps, } l \text{ - steps, with } m - l = k,\ m + l = n)$$

Now let's look at how we can use this to derive an equation for $P(X_n)$, and to derive a partial differential equation for this probability in the continuum limit - that is the limit as we "zoom out", so steps and time intervals shrink.

space ↘ ↙ time

Random walk: use the shorthand, $P(k, j) = P(X_j = k)$

$$\begin{aligned} P(k,j) &= P(X_j = k) = pP(X_{j-1} = k-1) + (1-p)P(X_{j-1} = k+1) \\ &= pP(k-1, j-1) + (1-p)P(k+1, j-1) \end{aligned}$$

For $p = \frac{1}{2}$ (symmetric) we rewrite this as:

$$\begin{aligned} P(k,j) - P(k,j-1) &= \frac{1}{2}[P(k-1,j-1) + P(k+1,j-1) - \\ &\quad 2P(k,j-1)] \end{aligned}$$

It doesn't matter what the grid size is, so, take time step size Δt, space step size ΔX,

$$P(X_k, t_j) - P(X_k, t_j - \Delta t) = \frac{1}{2}[P(X_k - \Delta X, t_j - \Delta t) + P(X_k + \Delta X, t_j - \Delta t) - 2P(X_k, t_j - \Delta t)]$$

Suppose

$$(\Delta X)^2 = C\Delta t$$

] diffusion scaling relationship

Then the equation for $P(X_k, t_j)$ becomes (for all k):

$$\frac{P(X_k, t_j) - P(X_k, t_j - \Delta t)}{C\Delta t} = \frac{1}{2(\Delta X)^2}[P(X_k - \Delta X, t_j - \Delta t) + \quad (57)$$

$$P(X_k + \Delta X, t_j - \Delta t) - 2P(X_k, t_j - \Delta t)] \quad (58)$$

heat/diffusion eq'n $\Longrightarrow P_t = \frac{C}{2}P_{XX}$ as $\Delta t, (\Delta x)^2 \to 0$ (59)

Exercise: Show how to obtain (59) via a Taylor series expansion about $\Delta t = 0$ and $\Delta X = 0$, with relationship $\Delta t \propto (\Delta X)^2$.

Use: $P(\Sigma_k \pm \Delta\Sigma, t_j - \Delta t) \sim P(\Sigma_k, \cdots) \pm P_\Sigma(\Sigma_k, \cdots)\Delta\Sigma + P_{\Sigma\Sigma}(\Sigma_k, \cdots)(\Delta\Sigma)^2/2 \cdots$

This PDE is the diffusion equation or the heat equation.

The solution is a mean zero Gaussian with variance proportional to t! This is the density of Brownian motion.

Soln to (59):

$$P(X,t) = \frac{1}{\sqrt{\pi C t}} e^{X^2/(Ct)} \quad \text{for } P(X,0) = \delta(X) \quad (60)$$

$$C/2 = \text{diffusion coefficient} \quad (61)$$

Exercise: Show that (60) is the solution to (59).
So we get Brownian motion under a diffusive scaling for the random walk.

This also illustrates the self-similarity of this diffusion. In particular, if $w(t)$ is a standard Brownian motion, then so is

$$\frac{1}{\sqrt{\alpha}} w(\alpha t), \qquad \langle (\frac{1}{\sqrt{\alpha}} w(\alpha t))^2 \rangle = t$$

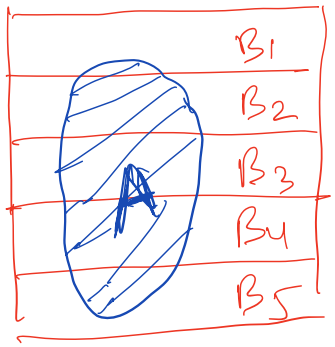
Fact from conditional probability:

$P(A \text{ occurs}) = P(A \text{ occurs, anything else occurs})$

We divide up "anything else" into N disjoint events, $B_1, B_2 \cdots B_N$

$\Rightarrow P(A, \text{any of } B_j \text{ occurs})$

$= \sum_{j=1}^{N} P(A, B_j) = \sum_{j=1}^{N} P(A|B_j)P(B_j)$



choose events that for which we know the conditional probability

Relationship between normal r.v.'s and Poisson r.v.'s

In your homework:

$X \sim$ Poisson r.v. w/ parameter λ

$$\Rightarrow f_X(x) = \frac{e^{-\lambda}\lambda^x}{x!} \quad x = 0, 1, 2 \cdots$$

X typically associated w/ # of events in a time interval, e.g. customers arriving to a store, # of events occuring in a day for a certain type of event

Poisson r.v. have an underlying assumption that the events occur independently

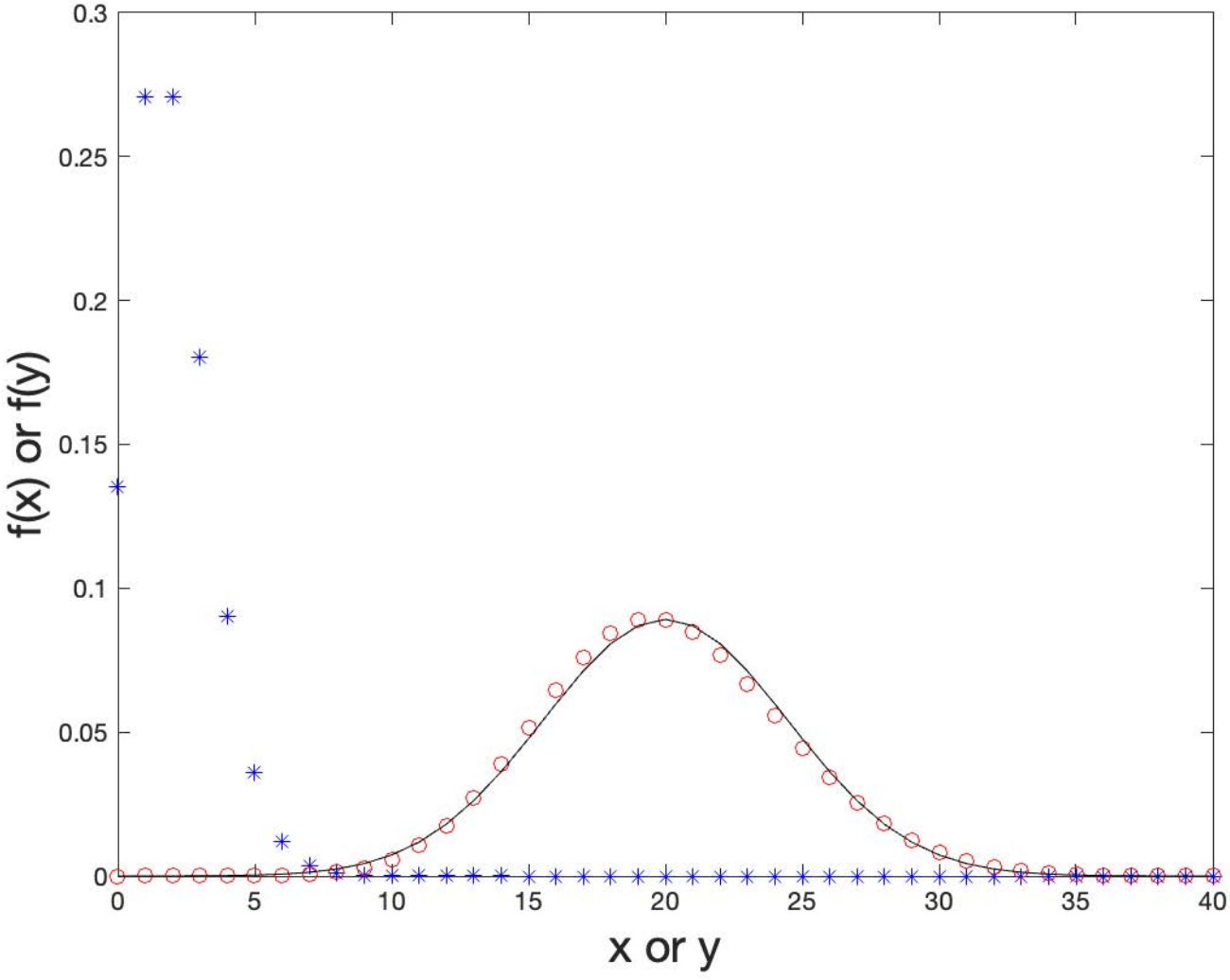
We often write events occurring using λ as a rate per time, then $X \sim$ Poisson w/ parameter $\lambda \Delta t$ for Δt a time interval

$$f_X(x) = \frac{e^{-\lambda \Delta t}(\lambda \Delta t)^x}{x!} \quad x = 0, 1, 2 \cdots$$

Since events are independent, then also # of events per time interval also independent, and we can use superposition of Poisson arrivals (events) to look at longer intervals, i.e. # of arrivals in an interval of length $N\Delta t$ is

$Y = \sum_{i=1}^{N} X_i$; Y is Poisson w/ parameter $\lambda N \Delta t$

$$f_Y(y) = \frac{(\lambda N \Delta t)^y e^{-N\lambda\Delta t}}{y!} \quad y = 0, 1, 2 \cdots$$


0.3
0.25
0.2
0.15
0.1
0.05
0
f(x) or f(y)
0
5
10
15
20
25
30
35
40
x or y

Compare the shape of $f_X(x)$ and $f_Y(y)$

For $\lambda = 2$, $\Delta t = 1$, $N = 10$

Not surprisingly, Y can be approximated w/ a normal r.v. ⇒ note Y is a sum of independent, identical r.v's X_i;

then, approximating Y as a normal, we use

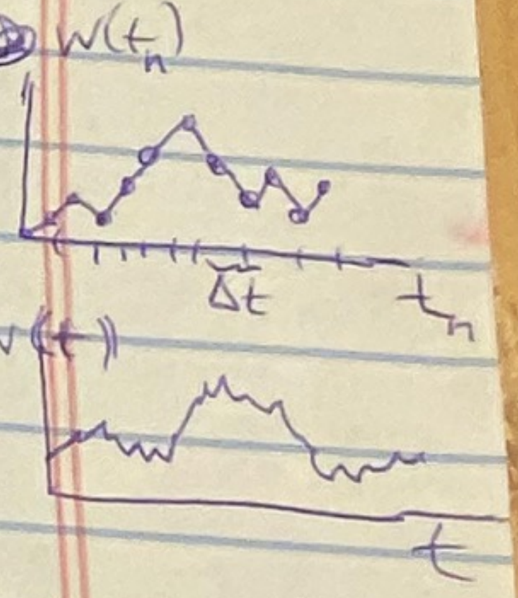
$E\left[\sum_{i=1}^{N} X_i\right] = N(\lambda \Delta t)$ $\quad$ $Var\left[\sum_{i=1}^{N} X_i\right] = N\lambda\Delta t$

(recall, $Var[X] = E[X]$ for X Poisson r.v.)

This framework allows us to consider more than one type of stochastic process

Diffusion: $W(t)$ obtained from a random walk, taking the limit of $\Delta t \to 0$, $\Delta x \to 0$ so that $(\Delta x)^2/\Delta t = \text{const} = D$

$W(t) \sim N(0, t)$ continuous w/ discontinuous derivative

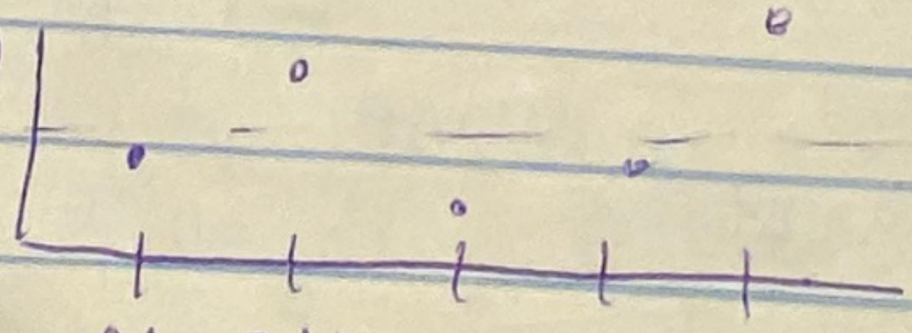


Poisson jump process:

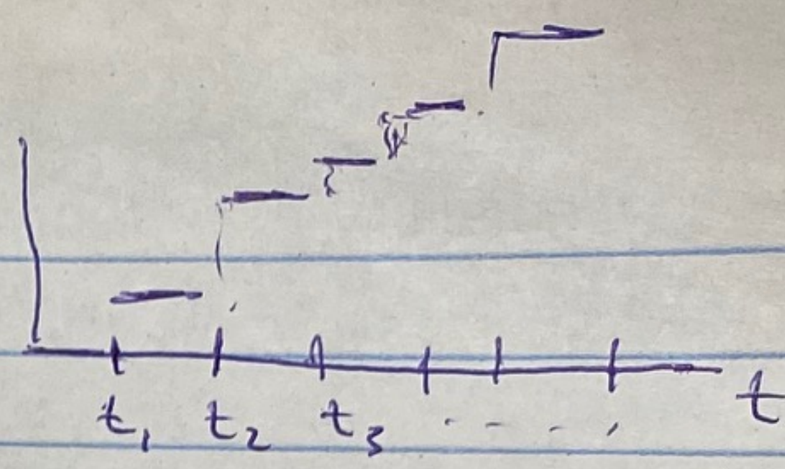
Consider $N(t)$ is total # of Poisson arrivals in a time interval $[0, t]$

Again consider $X(t) \sim Poiss(\lambda \Delta t)$, λ arrival rate per time unit,

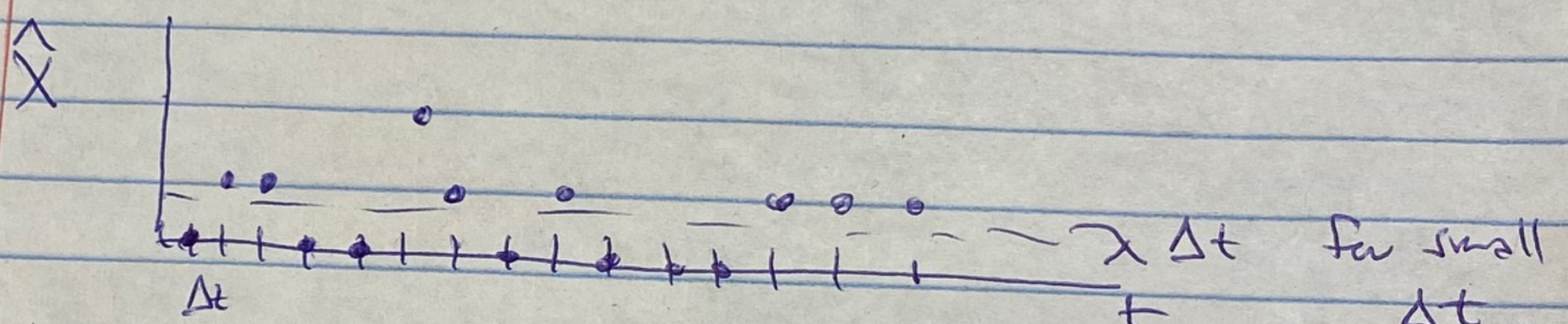
e.g.



Then $N(t) = \sum_{i=1}^{n} X_i(t_i)$



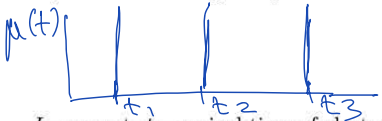
Note: if we take Δt small enough, then there may be some intervals w/ no arrivals, i.e. we divide up the interval $[0,t]$ into small intervals for observing # of arrivals, we may observe more often than the typical inter-arrival interval



how we choose this interval may depend on the application or the model we wish to study, i.e, do we want to observe individual arrivals, or is it OK to lump together?

Dynamics driven by Jump Processes - "Shot Noise"

(analogous to Ornstein-Uhlenbeck but w/ jump process)

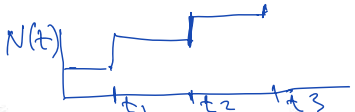


$$\frac{dI}{dt} = -\alpha I + q\mu(t) \tag{18}$$

$$\mu(t) = \sum \delta(t - t_k) \tag{19}$$

I=current, t_k=arrival time of electron. Note that $\mu(t)$ is not mean zero.

Define: $N(t)$=sum of Poisson arrivals, $\int_0^t dN = \int_0^t \mu\, dt = N(t)$



$$\frac{dN}{dt} = \mu \tag{20}$$

formally, but N is a step function. If we want to write the current equation in terms of a mean zero process (analogous to X in previous example) then consider $E[dN] = \lambda dt$, $Var[dN] = \lambda dt$, where λ is the mean arrival rate.

Note: For a Poisson process N with arrival rate λ, then the average number of arrivals in an interval of length t is λt.

Then

$$d\eta = dN - \lambda dt$$

is a centered Poisson increment, with mean zero, and variance λdt

Formally

mean increase in N

$[dI = (-\alpha I + \lambda q)\Delta t + q\, d\eta]$

$$\frac{dI}{dt} = -\alpha I + \lambda q + q\frac{d\eta}{dt} \tag{21}$$

} fluctuation around mean

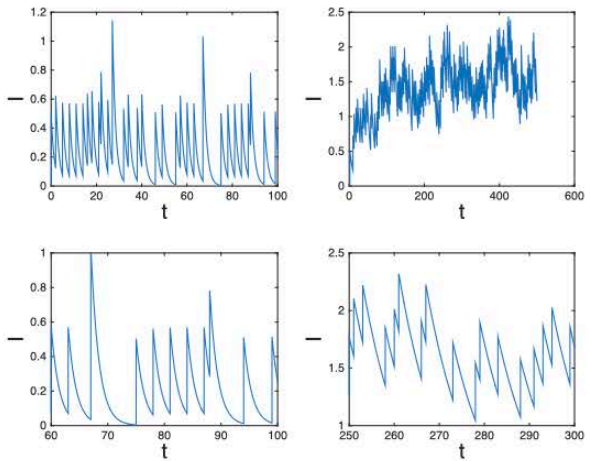


Figure 3: Realization for I. On the left, the rate of jumps is smaller compared with the decay rate of I, on the right the jumps are more frequent. The bottom row figures are zooms of the realizations in the top row.

just the "usual" ODE

Averaging yields

$$\frac{d\langle I\rangle}{dt} = (-\alpha\langle I\rangle + \lambda q), \quad \langle I\rangle \to \frac{\lambda q}{\alpha} \quad \text{as} \quad t \to \infty \tag{22}$$

What about $\langle I^2\rangle$?

(not zero, since N(t) additive)

Suppose we consider $d(I^2)$ and compare with equation above. We can write $d(I^2)$ in a couple of different ways:

$$d(I^2) = (I + dI)^2 - I^2 = 2IdI + (dI)^2 \tag{23}$$

and

$$(dI)^2 = ((-\alpha I + \lambda q)dt)^2 + 2(-\alpha I + \lambda q)dtd\eta + q^2 d\eta^2 \tag{24}$$

Compare with heuristic as in previous example: multiply equation (21) for I by I and integrate and average:

$$\frac{1}{2}\frac{d\langle I^2\rangle}{dt} = -\alpha\langle I^2\rangle + \lambda q\langle I\rangle + q\left\langle I\frac{d\eta}{dt}\right\rangle \tag{25}$$

What about $\left\langle I\frac{d\eta}{dt}\right\rangle$? Is it $= 0$? If so, then we can solve for $\langle I^2\rangle$ as $t \to \infty$ (left as an exercise).

$$\begin{aligned} \langle I^2\rangle &= \left(\frac{\lambda q}{\alpha}\right)^2 = \langle I\langle^2 \\ \Rightarrow Var(I) &= \langle I^2\rangle - \langle I\rangle^2 = 0 \end{aligned}$$

! No effect of noise - not correct!

This heuristic neglects the term $(dI)^2$ - it is not in the equation for $\langle I^2\rangle$, obtained from (23)-(24). So we need to examine that equation

$$(dI)^2 = (\quad)(dt)^2 + (\quad)dt\langle d\eta\rangle + q^2\langle d\eta^2\rangle \tag{26}$$

Note on Numerical simulation

Here η is not a continuous process, rather, events occur at times that are exponentially distributed. (later)

Then some time interval increments of length Δt will have an event, and some will not.

using the properties of the centered Poisson increment.

$$\langle d\eta \rangle = 0 \qquad \langle (d\eta)^2 \rangle = \lambda dt$$

Note that the centered Poisson increment $d\eta$ has the same variance as the original Poisson increment dN.

So now, combining results from the expression above, we can get the correct equation for I^2:

$$d\langle (I)^2 \rangle = 2\langle IdI \rangle + \langle (dI)^2 \rangle \quad (27)$$

$$= \langle (2I(-\alpha I + \lambda q))dt \rangle + \langle 2qIdN \rangle + q^2\lambda dt + o(dt) \quad (28)$$

use $\langle I \rangle$ from (27) $\Rightarrow \langle (I)^2 \rangle - \langle I \rangle^2 \Rightarrow Var\langle I \rangle = \frac{q^2\lambda}{2\alpha} \quad t \to \infty \quad (29)$

Recall that for the Wiener process $\langle w(t) \rangle = 0$ and $\langle w^2 \rangle = t$. Later we see that the if Poisson events are very frequent, then one can replace Poisson random variables with appropriate "centering" and Brownian motion (normal random variables).

Recall for Ornstein-Uhlenbeck process

$dV = -\alpha V + \beta dW$ as $t \to \infty$ $V \sim N(0, \sqrt{\frac{\beta^2}{2\alpha}})$

4 Approximations of interacting individuals

(using centered Poisson increments)

Models: spread of disease Susceptible population = S: don't have the disease
Infective population = I: has the disease
Population of recovered individuals R: had the disease, can't get it again

For either probabilistic or deterministic models, we think in terms of rates

transition rates tell us how we can increase or decrease

transition	rate
$S \to S+1$	μN
$S \to S-1$	$\beta SI/N + \mu S$
$I \to I+1$	$\beta SI/N$
$I \to I-1$	$(\gamma+\mu)I$
$R \to R+1$	γI
$R \to R-1$	μR

combination of 2 Poisson processes

$S+I+R = N$ = total population size: we will look at the case for large N, $N \gg 1$
μ = birth/death rate μ
γ = recovery rate
$\beta I/N$ = average number of contacts with infectives per susceptible per unit time.
N = total population size

Stochastic model: the rates (per unit time) are the conditional transition rates of the stochastic (Poisson) process (S, I, R)

Can write this as a Continuous time Markov process: $\{(S_t, I_t, R_t) : t \in [0, \infty)\}$, state space Z_+^3. N = expected population size $N = E[S + I + R]$

We need to break this down into Poisson events - births, deaths, infections, recoveries

For example: $P(S_{t+\Delta t} = s+1 | S_t = s) = \mu N \Delta t + o(\Delta t)$
A birth process is a Poisson event with probability $\mu N \Delta t$ for a birth in time interval of length Δt

Other events:
$P(S_{t+\Delta t} = s+1 | S_t = s) = \mu N \Delta t + o(\Delta t)$,
$P(S_{t+\Delta t} = s-1 | S_t = s) = \mu S \Delta t + o(\Delta t)$ decrease in susceptibles due to death
$P(S_{t+\Delta t} = s-1 | S_t = s) = \beta SI/N \Delta t + o(\Delta t)$ decrease in susceptibles due to infections
$P(I_{t+\Delta t} = i+1 | I_t = i) = \beta SI/N \Delta t + o(\Delta t)$, increase in infectives (only by infections)

use Δt small, $e^{-(\,)\Delta t} \approx 1$
$(\Delta t)^n = o(\Delta t)$ for $n > 1$ (neglect)

$P(S_{t+\Delta t} = s+1 | S_t = s) = P(1 \text{ increase})$
$+ P(2 \text{ increases}, 1 \text{ decrease}) + P(\text{other combinations to get } s+1)$
$= (\mu N \Delta t)^1 \frac{e^{-\mu N \Delta t}}{1!} \approx 1$ for small Δt
$+ (\mu N \Delta t)^2 \left(\frac{\beta SI}{N} + \mu S\right) \Delta t \frac{e^{-\mu N \Delta t - \left(\frac{\beta SI}{N} + \mu S\right)}}{2!\,1!}$
$+$ other terms $(\Delta t)^n$ $n > 1$

$(\Delta t)^n \cdot 1$ smaller than Δt for Δt small

Approximations

Deterministic (mean field) model

$$\frac{dS}{dt} = \mu(N - S) - \beta\frac{SI}{N},$$
$$\frac{dI}{dt} = \beta\frac{SI}{N} - (\gamma + \mu)I, \tag{72}$$

Note: R equation is redundant, $S + I + R = N$.

Basic reproductive number, $R_0 = \frac{\beta}{\mu+\gamma}$
If $R_0 > 1$ there is a stable endemic ($I \neq 0$) equilibrium. (ODE exercise)

We express the stochastic equations of the process in a form easily compared with the equations (72) of the deterministic model.

Recall diffusion processes - approximating a random walk with a Brownian motion for $(\Delta x)^2 = \Delta t$.

Ultimately, we will approximate Poisson process increments with Brownian motion increments.

It is useful to recall the centered Poisson increments that were considered in the model for shot noise. Then the centered increments will have mean zero and variance proportional to Δt.
Viewing our Poisson processes as increments: To each increment we add and subtract its conditional expectation, conditioned on the value of the process at the beginning of the time increment of length Δt. Each increment of the process is then the sum of the expected value of the increment of the process and the centered increment,

For a Poisson process U, composed of two independent types of Poisson events with probability P_+, P_- for increasing or decreasing by one, we have

$$P(U_{t+\Delta t} = u + 1 | U_t = u) = P_+\Delta t + o(\Delta t)$$
$$P(U_{t+\Delta t} = u - 1 | U_t = u) = P_-\Delta t + o(\Delta t)$$

Then the equation for an increment of U is written as the average change plus the increments:

$$\Delta U = (P_+ - P_-)\Delta t + Z_+ + Z_-,$$
$$Z_{+/-} = U^{+/-} - E[U^{+/-}]$$

Z_+ Z_- are centered Poisson increments, that is $E[Z_{+/-}] = 0$ with variances $P_{+/-}\Delta t$

Note: Can simplify, for two processes independent: $\Delta U = (P_+ - P_-)\Delta t + Z$
Z is a centered Poisson increment, that is $E[Z] = 0$ with variance that is the sum of the two variances $(P_+ + P_-)\Delta t$

Recall: σdW has distribution $N(0, \sigma\sqrt{\Delta t})$, so Z has zero mean and standard deviation proportional to $\sqrt{\Delta t}$, as does the standard Brownian motion.

When is a Poisson random variable well-approximated by a Gaussian random variable? Typically when there are enough events so that the Poisson parameter is large (e.g. > 10)

So we can write:

$$\Delta S = (\mu(N - S) - \beta\tfrac{SI}{N})\,\Delta t + \Delta Z_1 - \Delta Z_2,$$
$$\Delta I = (\beta\tfrac{SI}{N} - (\gamma + \mu)I)\,\Delta t + \Delta Z_2 - \Delta Z_3. \tag{73}$$

The increments ΔZ_1, ΔZ_2, ΔZ_3 are independent, centered Poisson variables with variances $\mu(N + S)\Delta t$, $\beta\frac{SI}{N}\Delta t$ and $(\gamma + \mu)I\Delta t$, respectively.

Note here that S and I have fluctuations that are not independent from each other. ΔZ_2 is the increment of the process that influences both S and I (infection), so it must appear in both equations. Thus the noise in the equations is correlated.

We replace ΔZ_i by increments of Brownian motion, dW_i, with the same standard deviations,

$$dS = \left(\mu(N - S) - \frac{\beta}{N}SI\right)\,dt + G_1 dW_1 - G_2 dW_2,$$
$$dI = \left(\frac{\beta}{N}SI - (\gamma + \mu)I\right)\,dt + G_2 dW_2 - G_3 dW_3, \tag{74}$$
$$G_1 = \sqrt{\mu(N + S)}, \quad G_2 = \sqrt{\frac{\beta}{N}SI}, \quad G_3 = \sqrt{(\gamma + \mu)I}.$$